The Unitary Singularity: A Formal Rejection of the Multiverse via Algorithmic Parsimony

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Abstract

The Many-Worlds Interpretation (MWI) of quantum mechanics and the Eternal Inflation model of cosmology both posit the existence of infinite, non-interacting parallel universes. We subject these hypotheses to three independent formal analyses: (1) Kolmogorov complexity, demonstrating that the algorithmic description length of a single universe with non-deterministic observers is exponentially shorter than that of infinite branching histories; (2) the Principle of Least Action, showing that universal branching at every quantum event violates the variational principle that governs all known physics; and (3) information-theoretic testability, proving that causally disconnected universes are formally indistinguishable from noise under any physically realizable measurement protocol. We conclude that the multiverse hypothesis fails the Principle of Computational Parsimony—it is not *wrong* in the sense of being contradicted by evidence, but rather *unfounded* in the sense that it is algorithmically equivalent to invoking an infinite number of unobservable entities to explain a finite set of observations. A single deterministic universe with non-deterministic observers (the "Unitary Singularity") is the maximally parsimonious ontology consistent with all known physics.

Keywords: multiverse, Many-Worlds Interpretation, Kolmogorov complexity, algorithmic parsimony, fine-tuning, Occam's razor, quantum foundations.

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1 Introduction

These foundational principles operate on established invariants Penrose [1989], Shannon [1948b].

The multiverse hypothesis—the claim that our universe is one of infinitely many—has become a prominent feature of contemporary physics and cosmology, motivated primarily by:

[label=(a)]

1. The **Fine-Tuning Problem**: The fundamental constants of physics appear tuned to permit complex structure and life [Barrow and Tipler, 1986, Rees, 2000].
2. The **Measurement Problem** in quantum mechanics: The MWI [Everett, 1957] avoids wavefunction collapse by postulating that all branches of the state vector are equally real.
3. **Eternal Inflation**: Inflationary cosmology generically produces pocket universes with varying physical constants [Guth, 1981, Linde, 1986].

While these motivations are substantive, the multiverse hypothesis introduces an extraordinary ontological commitment: the existence of infinitely many unobservable entities. We argue that this commitment is not justified by the explanatory work it performs.

1.1 Main Results

1. **Kolmogorov Parsimony** (§2): The algorithmic description of a branching multiverse has Kolmogorov complexity $(M) \to \infty$, whereas the single-universe model has finite $(U) < \infty$.
2. **Variational Violation** (§3): Universal branching at every quantum event requires infinite energy and storage, violating the Principle of Least Action.
3. **Information-Theoretic Untestability** (§4): Causally disconnected universes carry zero mutual information with our observational data; they are formally noise.
4. **The Unitary Alternative** (§5): A single deterministic universe with a non-deterministic observer is sufficient to explain all known phenomena, including fine-tuning and quantum indeterminacy.

2 The Kolmogorov Argument

Definition 2.1 (Kolmogorov Complexity)**.** The Kolmogorov complexity (x) of a string x is the length of the shortest program that produces x on a universal Turing machine [Kolmogorov, 1965, Li and Vitányi, 2008]:

$$(x) = \min_p \{|p| : U(p) = x\}. \tag{1}$$

Theorem 2.2 (Parsimony Inequality)**.** *Let U denote the description of a single deterministic universe with n observer-induced non-deterministic measurements, and let M denote the description of a multiverse with branching at each measurement. Then:*

$$(U) = (\textit{laws}) + O(\log n) \quad \ll \quad (M) = (\textit{laws}) + O(2^n), \tag{2}$$

since M must specify the state of 2^n branches while U specifies only the observed branch.

Proof. The single-universe model requires: (i) the physical laws (finite description), and (ii) the outcomes of n quantum measurements (at most n bits, representable in $O(\log n)$ additional description via compression of random sequences). The multiverse model requires: (i) the same physical laws, plus (ii) the complete state specification of all 2^n branches. Since each branch is a full universe-state, $(M) \geq (\text{laws}) + 2^n \cdot (\text{state per branch})$. The inequality follows. $\square$

Corollary 2.3 (Occam's Razor, Formalized)**.** *By Solomonoff's theory of inductive inference [Solomonoff, 1964], the prior probability of a hypothesis h is $P(h) \sim 2^{-(h)}$. Therefore:* $P(U)/P(M) \sim 2^{(M)-(U)} \to \infty$.

3 The Variational Argument

[Principle of Least Action] All known fundamental physics obeys the Principle of Least Action: the path taken by a system between two states is the one that extremizes the action $S = \int \mathcal{L}\, dt$ [Feynman, 1942, Landau and Lifshitz, 1976].

[Branching Violates Parsimony of Action] The MWI requires that at every quantum measurement, the *entire universe* duplicates into k branches (where k is the dimension of the measurement basis). The total action of the multiverse after n measurements is:

$$S_{\text{total}} = \sum_{i=1}^{k^n} S_i, \tag{3}$$

where S_i is the action of the i-th branch. This sum grows exponentially, whereas the single-universe action S_U is bounded. A system that minimizes action cannot also maximize the number of copies of itself.

Proponents of MWI argue that branches do not require "additional energy" because the total wavefunction is unitary. However, unitarity governs the *norm* of the state vector, not its *ontological commitment*. The question is not whether the mathematics is consistent but whether the ontology is parsimonious.

4 Information-Theoretic Testability

Definition 4.1 (Mutual Information)**.** The mutual information between two random variables X (our universe's data) and Y (a parallel universe's data) is:

$$I(X;Y) = H(X) + H(Y) - H(X,Y), \tag{4}$$

where H denotes Shannon entropy [Shannon, 1948a, Cover and Thomas, 2006].

Theorem 4.2 (The Noise Theorem)**.** *If universe Y is causally disconnected from universe X (i.e., no physical signal can travel between them), then X and Y are statistically independent, and:*

$$I(X;Y) = 0. \tag{5}$$

A hypothesis that predicts entities with zero mutual information with all observational data is empirically indistinguishable from a hypothesis that predicts no such entities.

Proof. Causal disconnection implies $P(X,Y) = P(X)P(Y)$. Then $H(X,Y) = H(X) + H(Y)$, and $I(X;Y) = 0$. By the likelihood principle, any observational data D satisfies $P(D \mid \text{multiverse}) = P(D \mid \text{single universe})$, rendering the multiverse unfalsifiable. □

5 The Unitary Singularity

Definition 5.1 (Unitary Singularity)**.** The *Unitary Singularity* is the ontological position that:

[label=(i)]

1. There exists exactly one physical universe.
2. Its global dynamics are deterministic (governed by unitary evolution of the state vector).
3. Local non-determinism arises from the observer's epistemic limitations: incomplete knowledge of initial conditions, not ontological branching.

Theorem 5.2 (Sufficiency of the Unitary Singularity)**.** *The Unitary Singularity accounts for all empirically observed phenomena that motivate the multiverse:*

[label=(b)]

1. ***Fine-tuning:*** *Explained by the non-invertibility of the causal generator function $f : \mathcal{G} \to \mathcal{O}$, where f^{-1} is undefined from within the observable frame. The constants are not "selected" from a distribution but are structural features of a unique system.*
2. ***Quantum indeterminacy:*** *Born-rule probabilities arise from the observer's position within the system (self-locating uncertainty), not from ontological branching.*
3. ***Wavefunction collapse:*** *The "collapse" is an update of the observer's state, not a physical process in the universe.*

6 Discussion

6.1 Comparison with Existing Critiques

Our analysis complements but extends existing critiques of the multiverse:

Critique	**Author(s)**	**Our Extension**
Unfalsifiability	Popper (1959)	Formalized via $I(X;Y) = 0$
Ontological excess	Occam/Lewis	Quantified via $(M) \gg (U)$
Action violation	— (novel)	Proved via variational principle
Observer sufficiency	Fuchs (QBism)	Unified with parsimony bounds

6.2 Limitations

1. Our Kolmogorov analysis assumes a fixed universal Turing machine; the complexity values are machine-dependent up to an additive constant.
2. The variational argument addresses ontological cost, not mathematical consistency—MWI remains internally consistent.
3. The fine-tuning resolution via non-invertibility is an axiom, not a derivation.

7 Conclusion

We have subjected the multiverse hypothesis to three independent formal tests:

1. **Algorithmic parsimony:** $(U) \ll (M)$.

2. **Variational principle:** Branching maximizes total action.

3. **Information content:** $I(X;Y) = 0$ for disconnected universes.

In each case, the multiverse fails: it is less parsimonious, violates the variational principle, and carries zero information content beyond what a single universe provides. The Unitary Singularity—one deterministic universe with epistemic non-determinism—is the simplest ontology consistent with all known physics.

The multiverse is not the "other worlds." It is the garbage collection of the wavefunction: variables that were considered but never executed.

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